MINI PROJECT: INTELLIGENT CONTROL OF DC MOTOR USING COMPUTATIONAL INTELLIGENCE TECHNIQUES

By Group 3

OUR TEAM

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PROJECT OBJECTIVES

1. To develop and simulate an intelligent controller for a DC motor.

2. To implement Computational Intelligence methods such as:
   - Fuzzy Logic
   - Genetic Algorithm
   - Artificial Neural Network
   - Hybrid Aproaches

3. To compare controller performance with or without intelligent control.

BASICS

J = 0.01 Moment of inertia (kg.m^2)

b = 0.1 Motor viscous friction constant (N.m.s)

K = 0.01 Electromotive force constant (V/rad/sec)

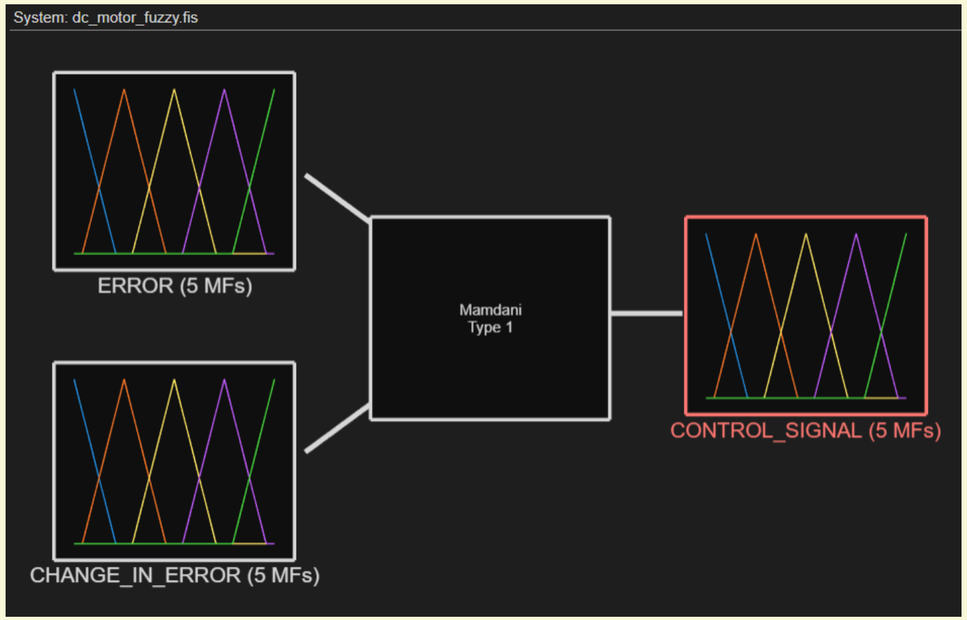
R = 1 Electric resistance (Ohm)

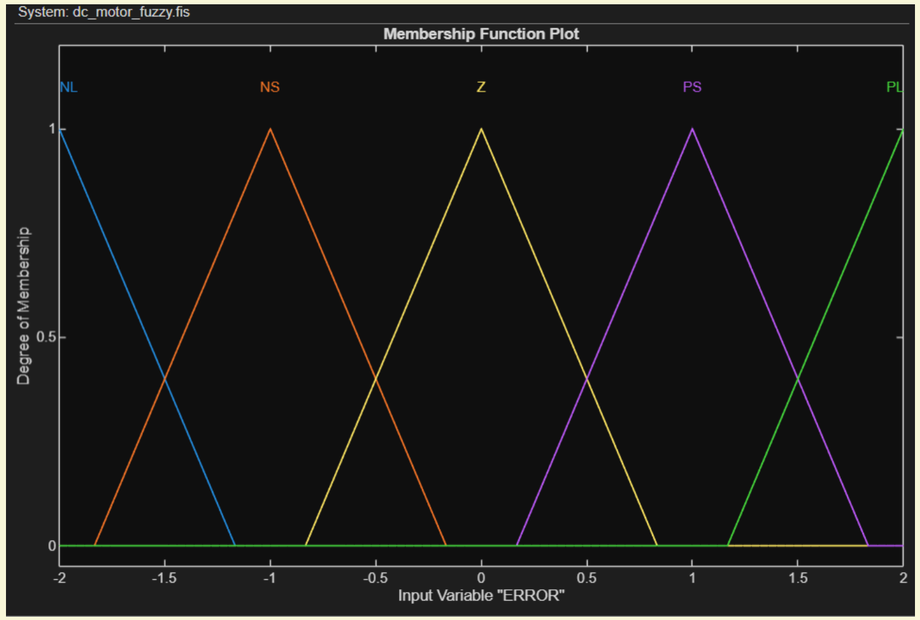
L = 0.5 Electric inductance (H)

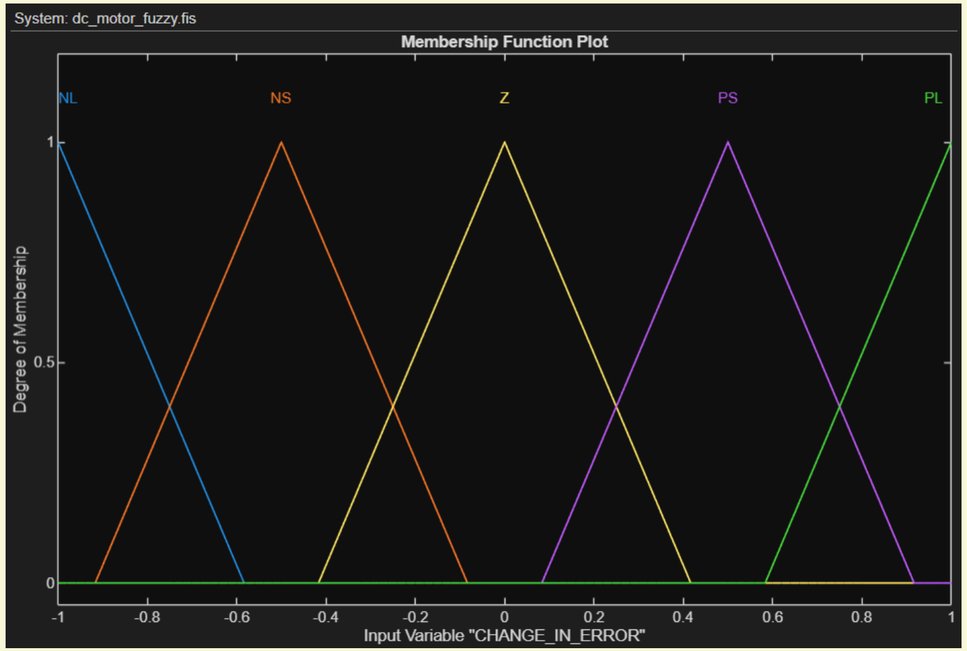
Made with MATLAB

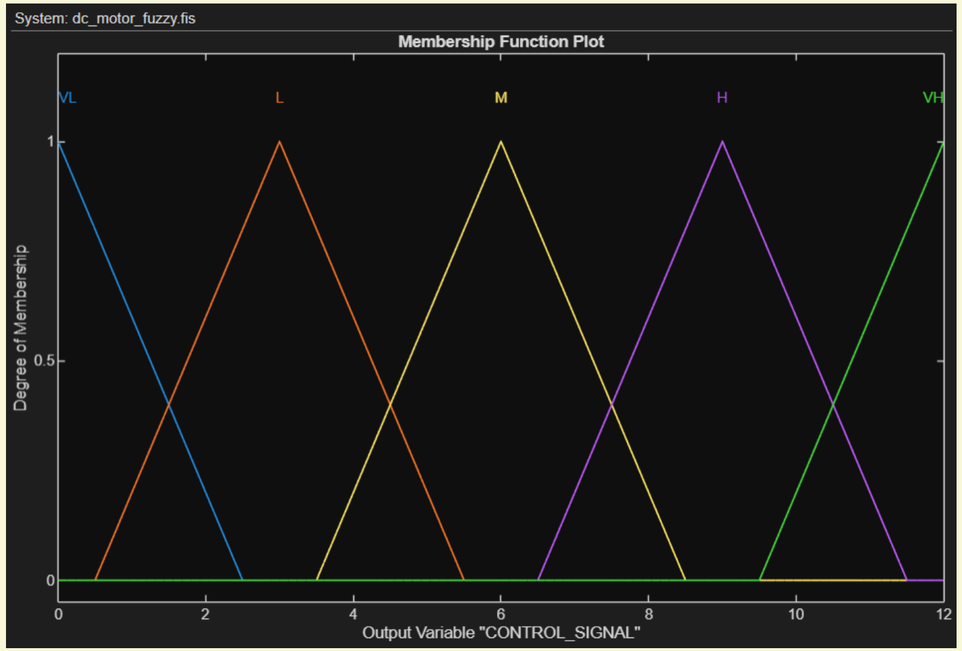
```
>> % DC Motor Parameters
J = 0.01;   % Moment of inertia (kg.m^2)
b = 0.1;    % Motor viscous friction constant (N.m.s)
K = 0.01;   % Electromotive force constant (V/rad/sec)
R = 1;      % Electric resistance (Ohm)
L = 0.5;    % Electric inductance (H)

% Transfer function
num = K;
den = [L*J, (L*b + R*J), (R*b + K^2)];
sys = tf(num, den);
```

$$G(s) = \frac{K}{LJs^2 + (Lb + RJ)s + (Rb + K^2)}$$


System: dc_motor_fuzzy.fis
ERROR (5 MFs)
CHANGE_IN_ERROR (5 MFs)
Mamdani
Type 1
CONTROL_SIGNAL (5 MFs)

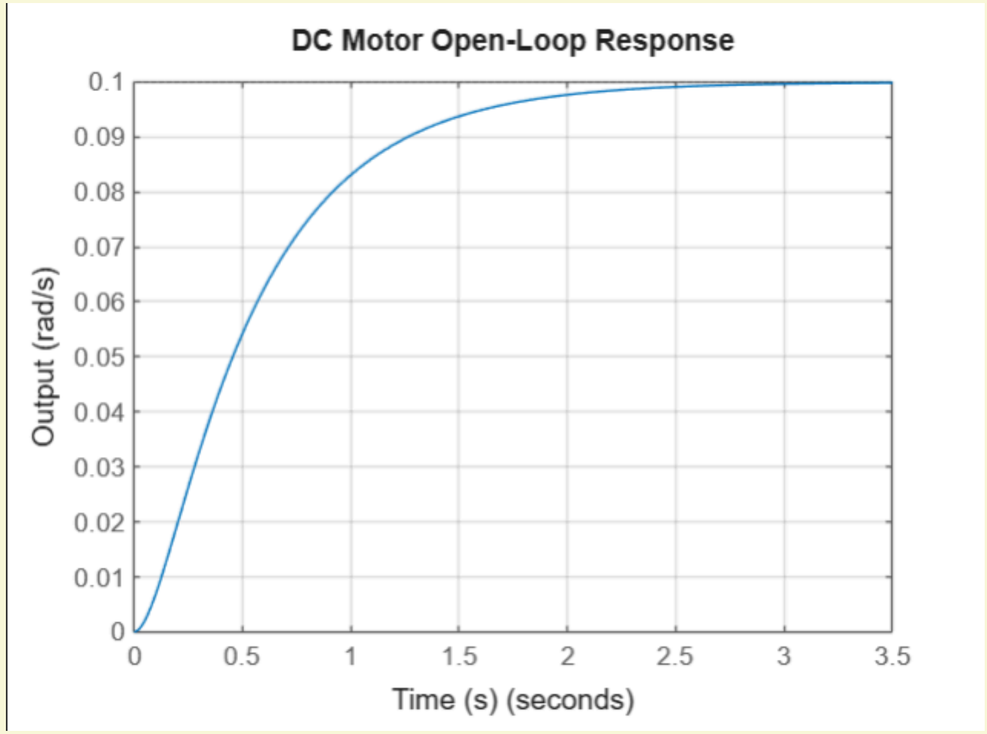

System: dc_motor_fuzzy.fis
Membership Function Plot
NL
NS
Z
PS
PL
Degree of Membership
1
0.5
0
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
Input Variable "ERROR"

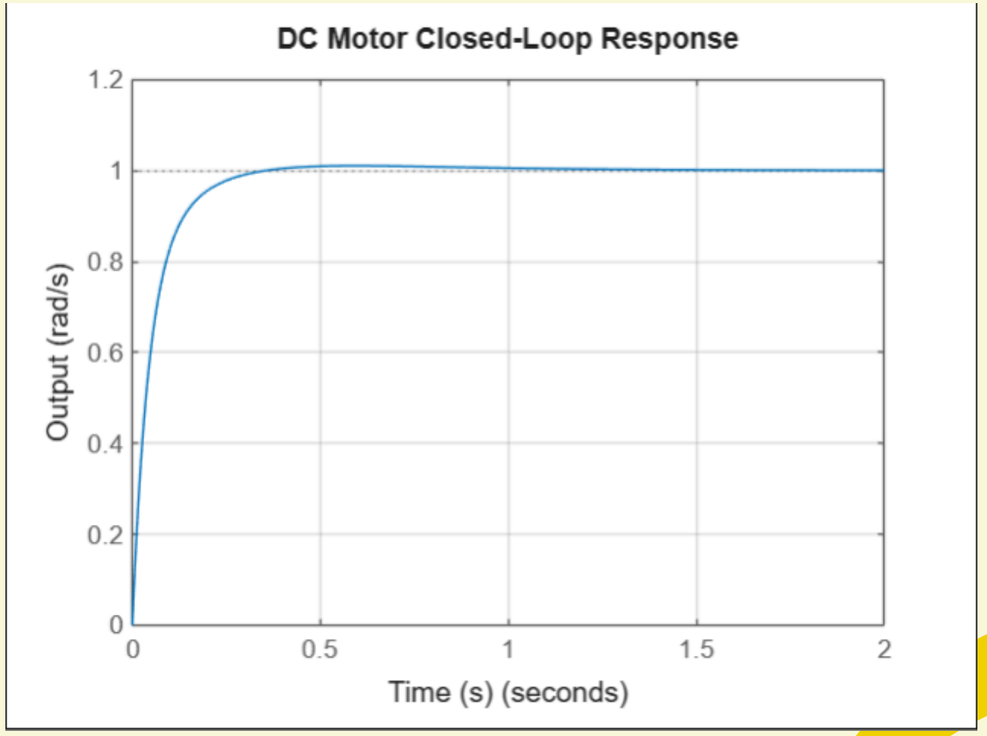

System: dc_motor_fuzzy.fis
Membership Function Plot
NL
NS
Z
PS
PL
Degree of Membership
1
0.5
0
-1
-0.8
-0.6
-0.4
-0.2
0
0.2
0.4
0.6
0.8
1
Input Variable "CHANGE_IN_ERROR"


System: dc_motor_fuzzy.fis
Membership Function Plot
VL
L
M
H
VH
Degree of Membership
1
0.5
0
0
2
4
6
8
10
12
Output Variable "CONTROL_SIGNAL"

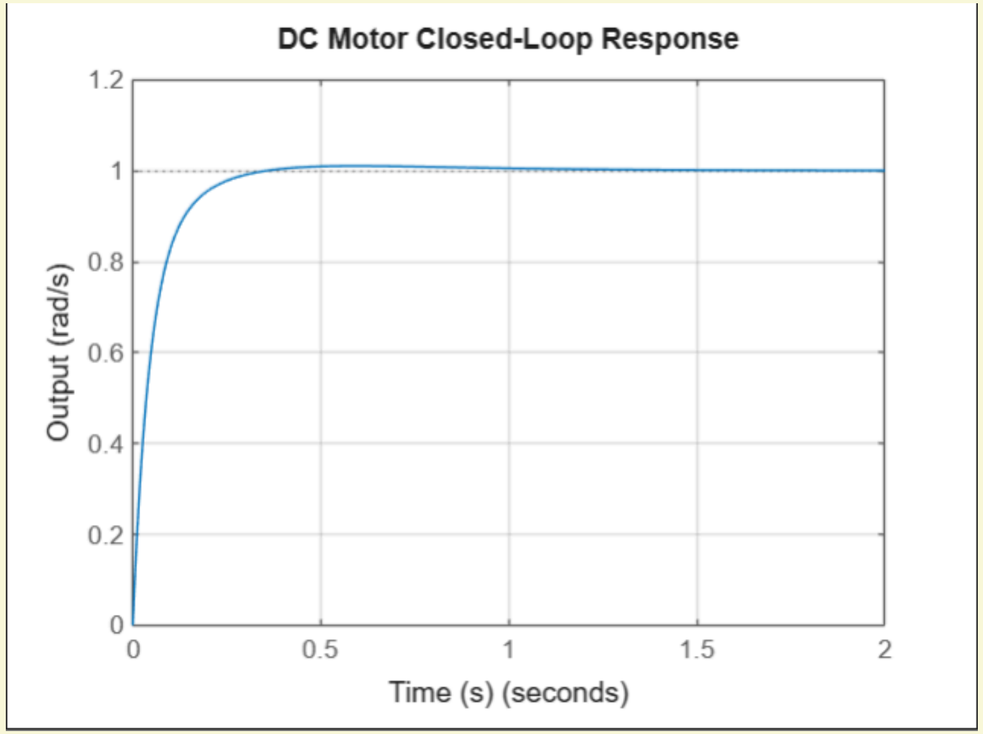
1	If ERROR is PL and CHANGE_IN_ERROR is NL then CONTROL_SIGNAL is VH
2	If ERROR is PL and CHANGE_IN_ERROR is NS then CONTROL_SIGNAL is VH
3	If ERROR is PL and CHANGE_IN_ERROR is Z then CONTROL_SIGNAL is H
4	If ERROR is PL and CHANGE_IN_ERROR is PS then CONTROL_SIGNAL is H
5	If ERROR is PL and CHANGE_IN_ERROR is PL then CONTROL_SIGNAL is M
6	If ERROR is PS and CHANGE_IN_ERROR is NL then CONTROL_SIGNAL is H
7	If ERROR is PS and CHANGE_IN_ERROR is NS then CONTROL_SIGNAL is H
8	If ERROR is PS and CHANGE_IN_ERROR is Z then CONTROL_SIGNAL is M
9	If ERROR is PS and CHANGE_IN_ERROR is PS then CONTROL_SIGNAL is M
10	If ERROR is PS and CHANGE_IN_ERROR is PL then CONTROL_SIGNAL is L
11	If ERROR is Z and CHANGE_IN_ERROR is NL then CONTROL_SIGNAL is M
12	If ERROR is Z and CHANGE_IN_ERROR is NS then CONTROL_SIGNAL is M
13	If ERROR is Z and CHANGE_IN_ERROR is Z then CONTROL_SIGNAL is M
14	If ERROR is Z and CHANGE_IN_ERROR is PS then CONTROL_SIGNAL is M
15	If ERROR is Z and CHANGE_IN_ERROR is PL then CONTROL_SIGNAL is M
16	If ERROR is NS and CHANGE_IN_ERROR is NL then CONTROL_SIGNAL is L
17	If ERROR is NS and CHANGE_IN_ERROR is NS then CONTROL_SIGNAL is L
18	If ERROR is NS and CHANGE_IN_ERROR is Z then CONTROL_SIGNAL is L
19	If ERROR is NS and CHANGE_IN_ERROR is PS then CONTROL_SIGNAL is VL
20	If ERROR is NS and CHANGE_IN_ERROR is PL then CONTROL_SIGNAL is VL
21	If ERROR is NL and CHANGE_IN_ERROR is NL then CONTROL_SIGNAL is VL
22	If ERROR is NL and CHANGE_IN_ERROR is NS then CONTROL_SIGNAL is VL
23	If ERROR is NL and CHANGE_IN_ERROR is Z then CONTROL_SIGNAL is VL
24	If ERROR is NL and CHANGE_IN_ERROR is PS then CONTROL_SIGNAL is VL
25	If ERROR is NL and CHANGE_IN_ERROR is PL then CONTROL_SIGNAL is VL

PID MOTOR CONTROLLER



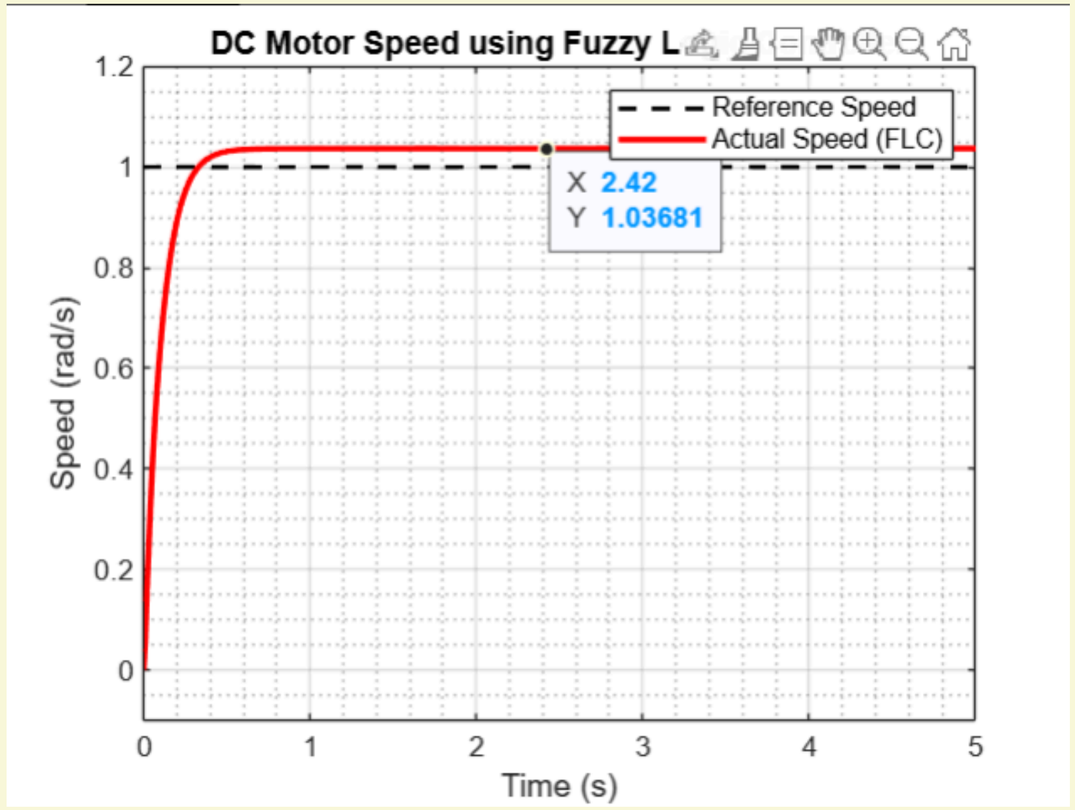


PID MOTOR CONTROLLER



- Very Fast Initial Acceleration
- 1.01% overshoot
- Minimal Error
- Takes 0.356s to reach setpoint
- Stabilizes at 1.03s
- Steady State Error around 0

FUZZY CONTROLLED



- Slower Initial Acceleration
- Some Error
- Takes 0.33s to reach setpoint
- Stabilizes at 1.16s
- Steady State Error = 0

COMPARE

PID

- Best Speed
- Sligth Overshooting
- Best Accuracy

FUZZY

- Most Stable
- Slower
- No Overshoot

CONCLUSION

Both options are viable in the industry. While PID is faster and more accurate it tends to overshoot slightly. On the other hand Fuzzy Logic Controlled is much more stable with no overshoot at the cost of lower speed and power.

QNA SESSION

THANK YOU